PG No. 33 $D_{2d}(1)$ $\bar{4}2m$ (-4m2 setting) [tetragonal]

Table 1: Wyckoff site: `1o`, site symmetry: `-4m2`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8]`

Table 2: Wyckoff site: `2a`, site symmetry: `2mm.`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,5,6]`
2	$[0, 0, -z]$	`[3,4,7,8]`

Table 3: Wyckoff site: `4b`, site symmetry: `..2`

No.	position	mapping
1	$[x, x, 0]$	`[1,7]`
2	$[-x, -x, 0]$	`[2,8]`
3	$[x, -x, 0]$	`[3,5]`
4	$[-x, x, 0]$	`[4,6]`

Table 4: Wyckoff site: `4c`, site symmetry: `.m.`

No.	position	mapping
1	$[x, 0, z]$	`[1,5]`
2	$[-x, 0, z]$	`[2,6]`
3	$[0, -x, -z]$	`[3,8]`
4	$[0, x, -z]$	`[4,7]`

Table 5: Wyckoff site: `8d`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[-x, -y, z]$	`[2]`
3	$[y, -x, -z]$	`[3]`
4	$[-y, x, -z]$	`[4]`
5	$[x, -y, z]$	`[5]`
6	$[-x, y, z]$	`[6]`
7	$[y, x, -z]$	`[7]`
8	$[-y, -x, -z]$	`[8]`